scientist of the twentieth century after Einstein, Paul Dirac, the great architect of quantum mechanics and author of the first and principal equations of the Standard Model, repeatedly expressed his dissatisfaction at this state of things, concluding that ‘we have not yet solved the problem’.

In addition, a striking limitation of the Standard Model has appeared in recent years. Around every galaxy astronomers observe a large cloud of material which reveals its existence via the gravitational pull that it exerts upon stars, and by the way it deflects light. But this great cloud, of which we observe the gravitational effects, cannot be seen directly and we do not know what it is made of. Numerous hypotheses have been proposed, none of which seem to work. It’s clear that there is *something* there, but we don’t know what. Nowadays it is called ‘dark matter’. Evidence indicates that it is something *not* described by the Standard Model, otherwise we would see it. Something other than atoms, neutrinos or photons …

It is hardly surprising that there are more things in heaven and Earth, dear reader, than have been dreamt of in our philosophy – or in our physics. We did not even suspect the existence of radio waves and neutrinos, which fill the universe, until recently. The Standard Model remains the best that we have when speaking today about the world of things, its predictions have all been confirmed; and apart from dark matter – and gravity as described in the theory of general relativity as the curvature of space-time – describes well every aspect of the perceived world.

Alternative theories have been proposed, only to be demolished by experiments. A fine theory proposed in the 1970s, and given the technical name SU5, for example, replaced the disordered equations of the Standard Model with a much simpler and more elegant structure. The theory predicted that a proton could disintegrate, with a certain probability, transforming into electrons and quarks. Large machines were constructed to observe protons disintegrating. Physicists dedicated their lives to the search for an observable proton disintegration. (You do not look at one proton at a time, because it takes too long to disintegrate. You take tonnes of water and surround it with sensitive detectors to observe the effects of disintegration.) But, alas, no proton was ever seen disintegrating. The beautiful theory, SU5, despite its considerable elegance, was not to the good Lord’s liking.

The story is perhaps repeating itself now with a group of theories known as ‘supersymmetric’, which predict the existence of a new class of particles. Throughout my career I have listened to colleagues awaiting with complete